

Designing a Responsible Multi-Agent Safety Architecture for Autonomous Vehicle Decision Systems

1. Executive Summary

This project presents a multi-agent safety architecture for autonomous vehicles (AVs) that embeds Responsible AI (RAI) principles directly into system design. The framework introduces a structured decision pipeline spanning pre-collision risk assessment, in-collision mitigation, post-event recovery, and post-hoc review. Unlike traditional black-box AV systems, this architecture emphasizes transparency, quantifiable risk evaluation, and auditability using real-world safety metrics.

2. Problem and Context

Autonomous vehicles must make high-stakes decisions under uncertainty. Current systems lack:

- Transparent decision reasoning
- Quantifiable risk modeling
- Clear accountability mechanisms

This project addresses these gaps by designing a system where safety, ethics, and governance are integrated into the architecture itself.

3. Multi-Agent Architecture Design

Core Agents

1. Perception Monitor Agent

2. Inputs: Sensor data (camera, LiDAR, radar)

3. Outputs: Object detection + uncertainty estimation

4. Risk Assessment Agent

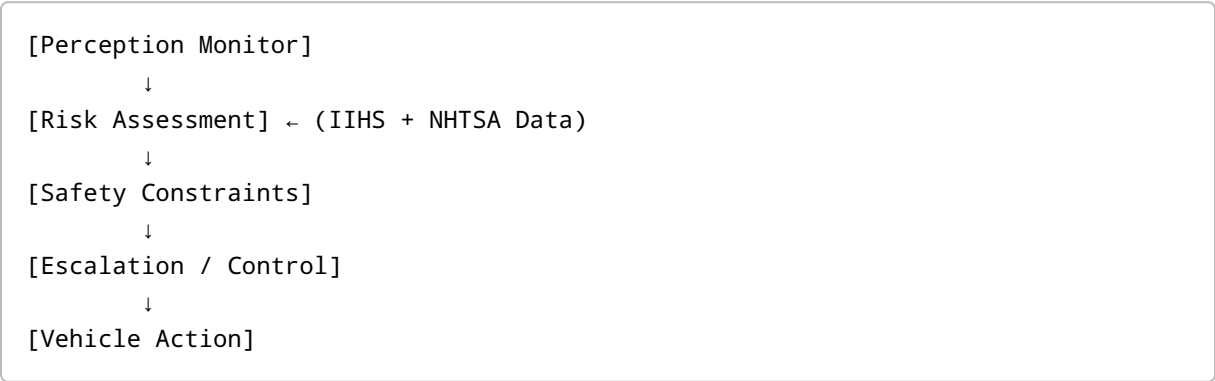
5. Computes probabilistic risk scores

6. Integrates real-world safety metrics (collision types, severity)

7. Safety Constraint Enforcement Agent

8. Applies ethical and safety constraints
 9. Optimizes decisions to minimize harm
 10. **Escalation / Override Agent**
 11. Triggers emergency actions or human takeover
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System Flow (Conceptual Diagram)



4. Temporal Safety Pipeline

The system operates across four phases:

4.1 Pre-Collision (Risk Assessment)

- Detect hazards
- Estimate uncertainty
- Compute risk scores

4.2 Collision Mitigation (During Event)

- Optimize trajectory
- Balance competing risks (pedestrian vs passenger)

4.3 Recovery (Post-Event)

- Stabilize vehicle
- Reassess environment
- Trigger emergency protocols

4.4 Review (Post-Hoc Analysis)

- Log decisions
- Enable auditability
- Support regulatory compliance

5. Risk Modeling Framework

Risk Function

Risk is modeled as:

$Risk = Probability \times Severity \times Exposure$

Extended Model with Uncertainty

$Adjusted\ Risk = Risk \times Uncertainty\ Factor$

Risk Components Table

Component	Description	Example Source
Probability	Likelihood of collision	Sensor prediction
Severity	Impact of collision	IIHS crash ratings
Exposure	Number/type of affected entities	Pedestrian density
Uncertainty	Confidence in perception	Sensor noise

6. Real-World Safety Integration

Data Sources

- IIHS crash safety ratings
- NHTSA collision statistics

Example Mapping

Collision Type	Severity Weight
Frontal	Medium
Side Impact	High

Collision Type	Severity Weight
Rollover	Very High

7. RAI Principle Mapping

Principle	Implementation Mechanism
Transparency	Decision logging
Accountability	Audit trails
Safety	Risk constraints
Human Oversight	Escalation agent

8. Scenario Analysis

Scenario: Pedestrian in Low Visibility

- Perception uncertainty increases
- Risk score inflated
- System prioritizes braking over speed maintenance

9. Risk Analysis

Technical Risks

- Sensor failure
- Model misclassification

Human Factors

- Trust in automation
- Delayed human intervention

Governance Risks

- Ambiguous accountability
- Lack of regulatory alignment

10. Governance and Oversight Design

- Logging of all decisions
 - Traceability of agent outputs
 - Compliance with regulatory frameworks
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11. Reflection

This architecture demonstrates how Responsible AI principles can be embedded structurally into AV systems. However, challenges remain in balancing modularity with accountability and ensuring real-world robustness.

12. Conclusion

The proposed system bridges the gap between ethical AI principles and real-time decision-making in autonomous vehicles by combining multi-agent design with quantifiable risk modeling and governance-aware architecture.